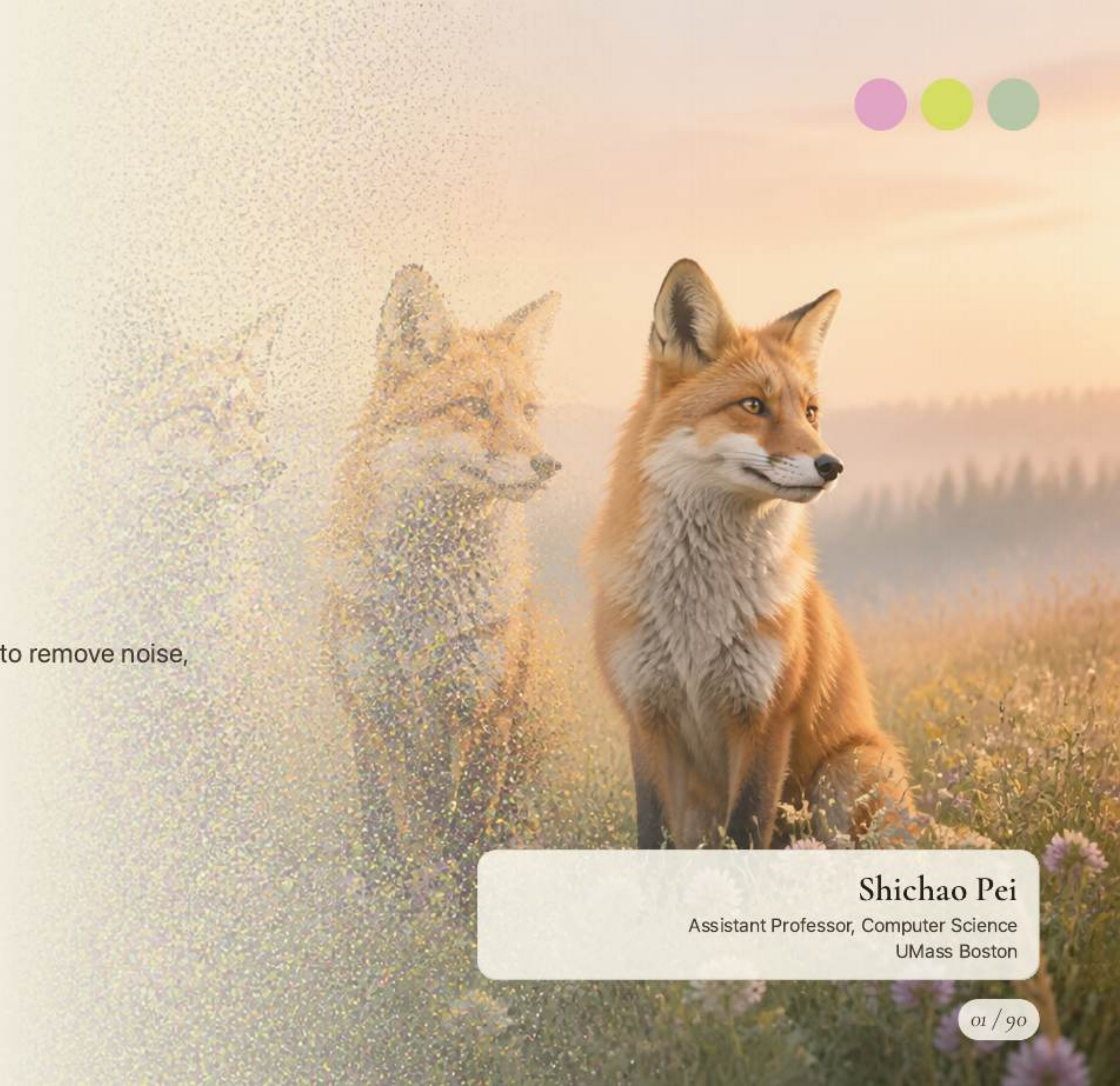




Day 2 · Image Generation

From Noise to Images.

How computers represent pictures, how diffusion models learn to remove noise, and how words guide the creation of something new.



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Three sections · ten modules

What we'll explore

We begin with pixels, follow the diffusion process, and finish by learning how prompts guide style, detail, and coherence.

Section 1

Images as data

- 01 Images are numbers
- 02 Color & resolution
- 03 Features & latent space

Section 2

Noise to image

- 04 The diffusion idea
- 05 Adding noise
- 06 Iterative denoising
- 07 Latent diffusion

Section 3

Words as guidance

- 08 Text conditioning
- 09 Prompt specificity
- 10 Coherence & responsible use

Yesterday predicted tokens. Today predicts visual change.

Both systems learn patterns from many examples, then make a sequence of small predictions.

Language generation

What token could come next?

An LLM repeatedly chooses a likely continuation.

text → token → token → token

Image generation

What visual change could help?

A diffusion model repeatedly predicts how to reduce noise.

noise → shape → detail → image

Before a model can create an image

Images are numbers

A computer receives a grid of numerical values—not a ready-made understanding of frogs, faces, or forests.



Start with the smallest visible unit

An image is a grid of pixels.

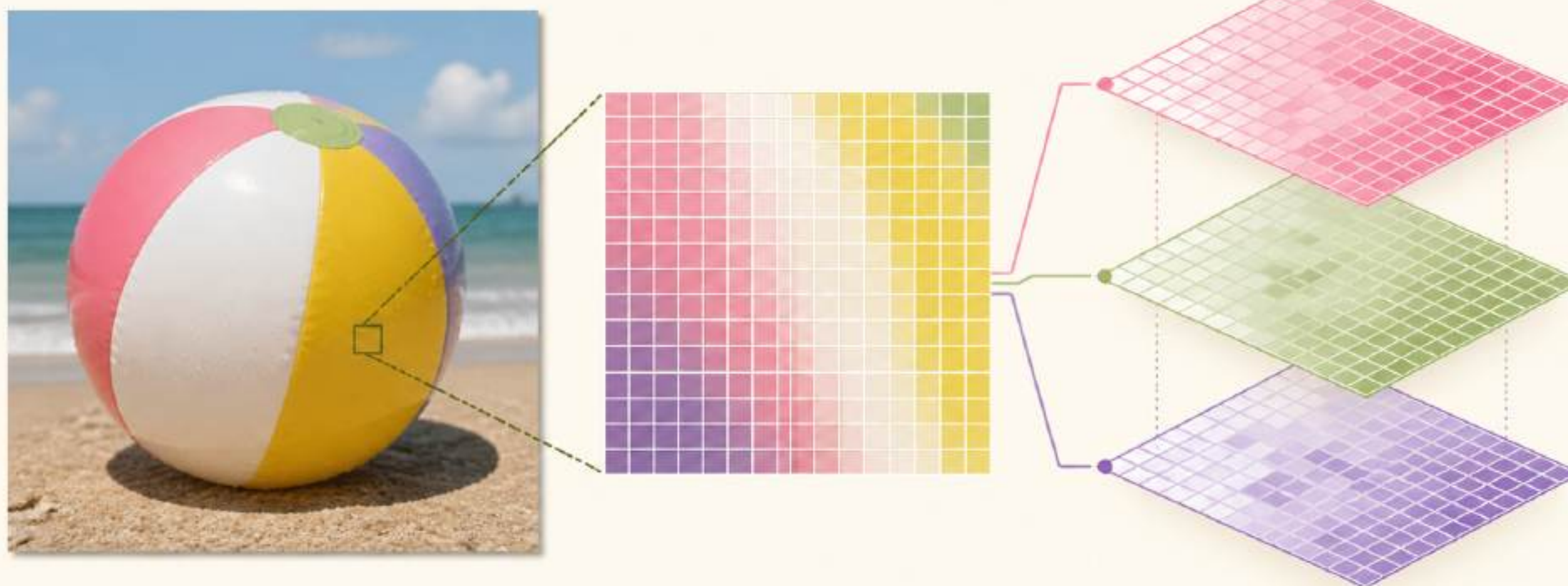
Each position in the grid stores information about brightness or color. Together, millions of small values produce the picture we see.

Pixel = picture element

A picture becomes structured data

The computer receives a block of values

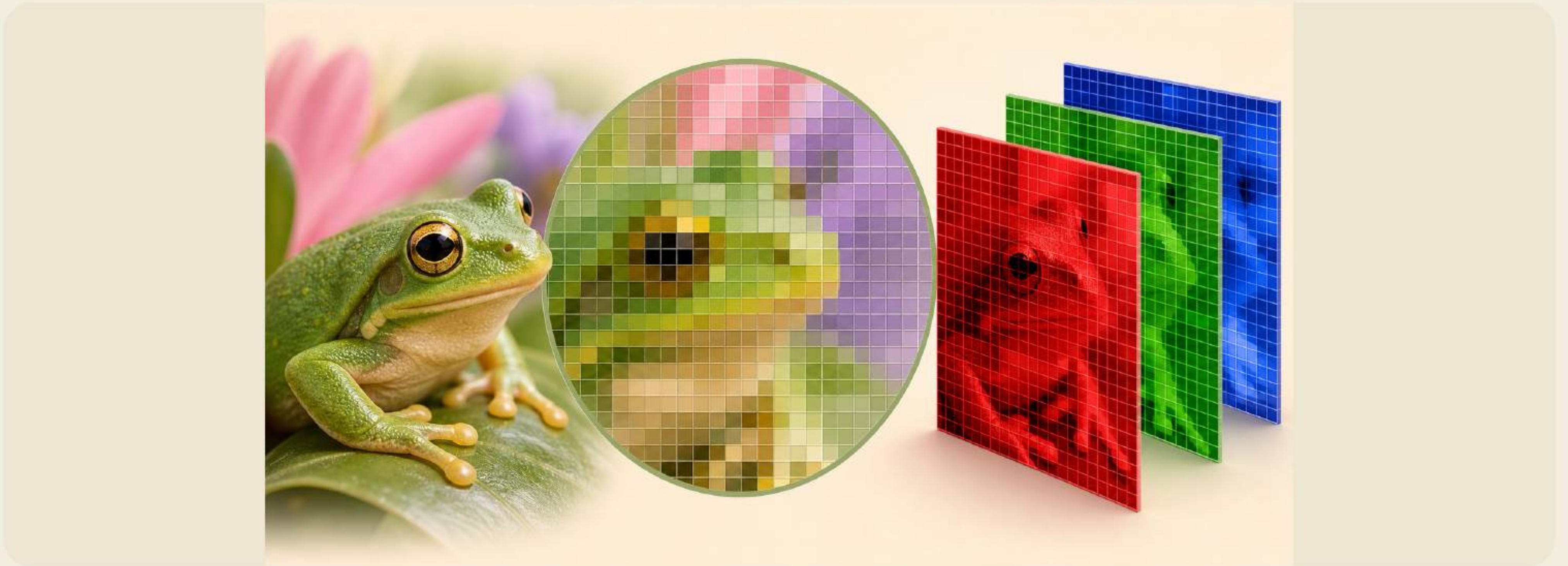
A photograph can be viewed as a pixel grid with aligned color planes: height $\times$ width $\times$ channels.



Zoom in on the data

One picture, many numerical pieces

A magnified region reveals the pixel grid. Color images often store separate red, green, and blue values.



A pixel value answers a small question

The exact format depends on the image, but the job is always local.

01

Where?

Its row and column identify a location.

02

How bright?

A grayscale pixel can store one intensity value.

03

What color?

A color pixel can store several channel values.

One value can control brightness

Dark values → bright values

A grayscale image may represent each pixel with a number along a brightness range.

black



white

Values between the endpoints create shades of gray.

One grid · sixty-four decisions

Can you decode the picture?

Read each value as a pixel instruction. Then compare the numerical grid with the image it creates.

Which rows reveal the shape first?

What changes if one value flips?

Would a larger grid preserve more detail?

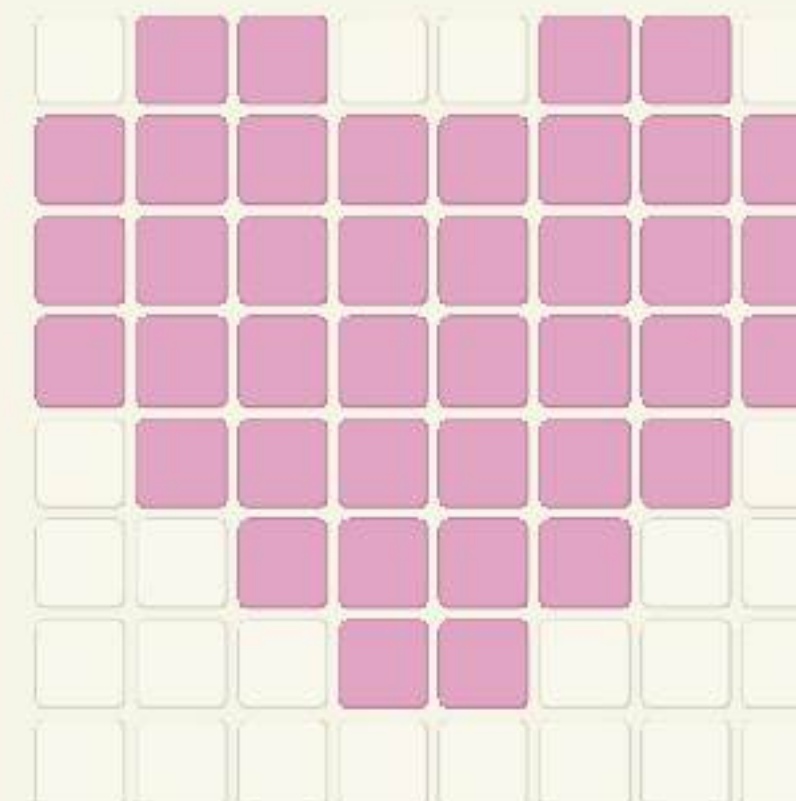
1 = filled pixel **0** = empty pixel

Step 1 · Read the values

0	1	1	0	0	1	1	0
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	0
0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0
0	0	0	0	0	0	0	0



Step 2 · See the image



A closer look at visual data

Color, channels & resolution

Color comes from combinations of channel values. Resolution controls how many pixel positions are available for detail.

Separate the signal · combine the result

RGB channels contribute different information

Each aligned channel records the intensity of one primary light color. Viewed together, the three maps reconstruct the full-color image.



Channel combinations create color

These examples are conceptual—the exact numerical scale can vary.

R

Mostly red

A strong red channel with weak green and blue can look red.

G

Mostly green

A strong green channel can create vivid greens.

B

Mostly blue

A strong blue channel can create deep blues and lilacs.

RGB

Together

Balanced high values can approach white light.

Three grids combine into one color image

Every location receives a red, green, and blue contribution.



Same scene · different grids

More pixels can preserve finer structure

Low resolution uses fewer positions, so small edges and textures may disappear or become blocky.



Width × height tells us how many pixel positions exist

A 1920 × 1080 image contains a grid with 1,920 columns and 1,080 rows.

Smaller grid

Less visual data

Faster to store and process, but fine detail may be lost.

fewer positions

Larger grid

More visual data

Can preserve more detail, but needs more memory and computation.

more positions

Detail is not free

Larger images require more work.

More pixels mean more values to store, compare, transform, and generate. This matters when an AI model repeats many operations.

More detail ↔ more computation

Same rocket · four pixel grids

When does the rocket become recognizable?

Move from large pixel blocks to a crisp image. The subject, framing, and colors stay the same; only the available detail changes.



Look from left to right

- 01 Which clue appears first: color or silhouette?
- 02 When do the window and fins become clear?
- 03 Does the final panel change the meaning—or only the detail?

Recognition often begins before every detail is visible.

Moving beyond raw pixel values

Features & latent representations

Models can transform detailed images into representations that emphasize useful patterns and reduce unnecessary complexity.



Pixels → edges → texture → parts → object

Small visual clues combine into larger ideas

The bicycle remains the same subject while each view highlights a different level of useful visual structure.



Color view → grayscale → edge map

Strong visual changes reveal boundaries

When neighboring values change sharply, the boundary may outline an object, a fold, a shadow, or a change in material.



Same shape · different surfaces

Surface patterns help describe materials

Smooth surfaces change gradually. Reflective, fuzzy, and rough surfaces create different local patterns—even when the overall shape stays the same.



No single clue carries the whole meaning

Models combine many weak signals across different locations and scales.

01

Shape

How boundaries form a silhouette.

02

Texture

How local patterns repeat.

03

Color

How channels vary across regions.

04

Context

How parts relate to the surrounding scene.

Keep what matters for the task

The model can describe an
image without copying every
pixel.

A learned representation may preserve patterns that help reconstruction or generation while
using a more compact internal form.

A compressed space of possibilities

Related images can have related representations.

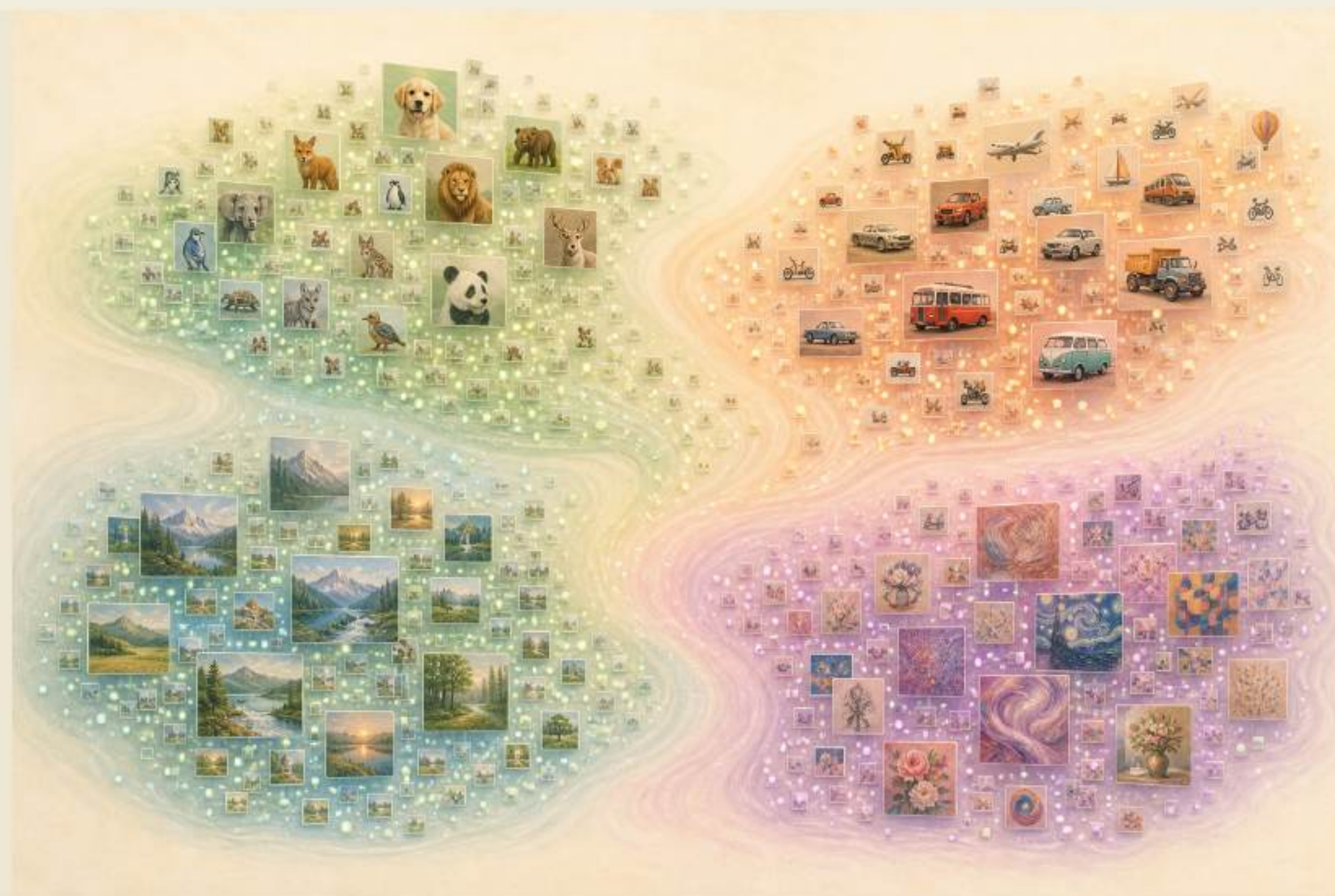
Nearby internal representations may share visual concepts, such as similar subjects, compositions, or styles.

Think of neighborhoods—not folders with perfect labels.

Conceptual latent neighborhoods

Patterns can cluster by visual similarity

Animals, vehicles, landscapes, and styles can form different neighborhoods. Real latent spaces have many dimensions, not just two.



Distance can reflect shared visual information

The geometry is learned from data and training—not designed as a perfect human dictionary.

Closer

More shared patterns

Two fox portraits may share subject, shape, and texture information.

Farther apart

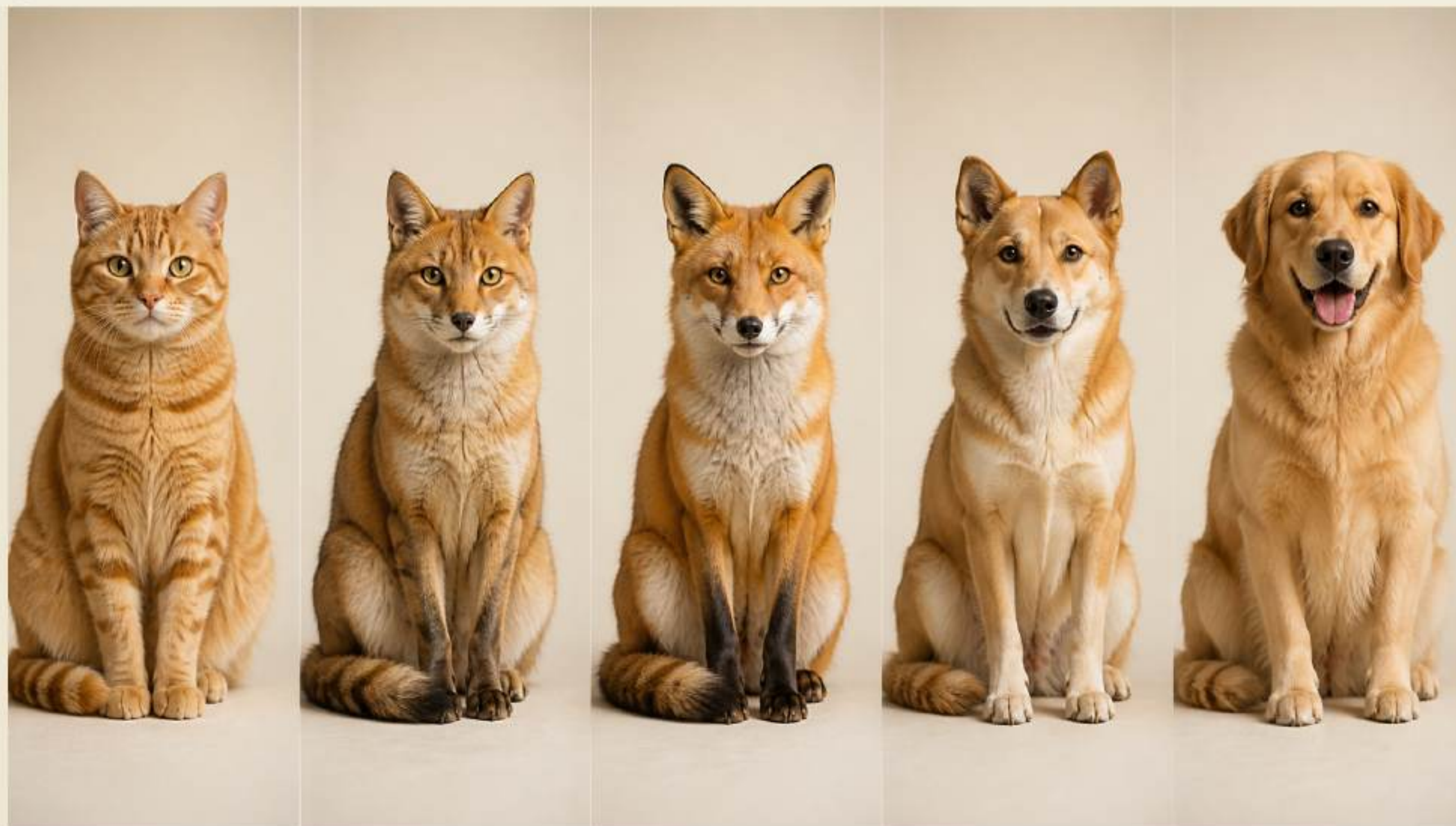
Fewer shared patterns

A fox portrait and a city blueprint may use very different features.

A conceptual interpolation

Small moves may create gradual visual changes

Intermediate representations can blend traits step by step. This cat-to-fox-to-dog path is an analogy, not a literal two-dimensional route inside every model.



Latent space is powerful—but not a literal map

The classroom picture simplifies a much more complex mathematical representation.

Helpful idea

Information is compressed and organized

Related visual patterns can influence nearby representations.

Important limit

There are no labeled streets inside the model

Real representations use many dimensions and learned numerical relationships.

Three ideas prepare us for diffusion

Now we can ask how an image model learns to move through visual representations.

01

Pixels

Images begin as grids of values.

02

Features

Patterns combine into useful structure.

03

Latents

Compact representations can reduce complexity.

The central mechanism

From noise to image

Diffusion models learn a gradual denoising process. Generation begins with uncertainty and builds structure through repeated predictions.

The big idea before the details

The diffusion idea

Instead of drawing a complete image in one jump, the model learns many smaller changes that gradually turn noise into coherent visual structure.

Generation as gradual refinement

A diffusion model learns to remove noise.

During training, the model practices predicting how a noisy example could become slightly cleaner. During generation, it repeats that skill many times.

Noise → prediction → slightly cleaner image

Clear examples make denoising practice possible

The model can compare its prediction with a known training example.

Known example

Start with a training image

A real image provides the clean target during learning.

Practice version

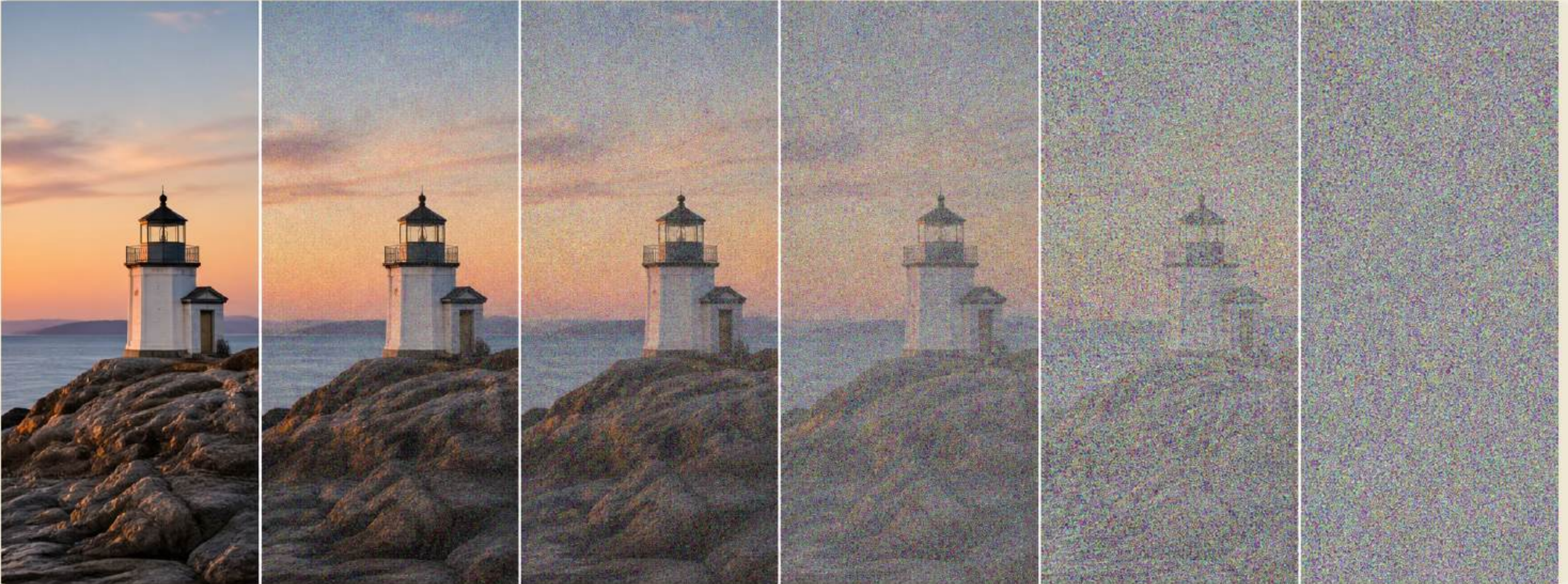
Add a controlled amount of noise

The model tries to predict the change needed to reduce it.

One lighthouse · increasing timesteps

Noise gradually hides recognizable structure

Color, edges, and silhouette weaken across the sequence until the original scene is nearly impossible to recover by inspection alone.



Smaller questions can be easier to learn

Many modest corrections can build a complex result.

At each step, the model solves a narrower prediction problem. The sequence turns a difficult image-generation task into repeated refinement.

Complex outcome = many simpler updates

The model does not perfectly reverse every random change

It learns statistical patterns that make cleaner images likely.

Exact reversal

Recover every original pixel

Usually impossible once strong random noise has erased information.

Learned denoising

Predict a plausible cleaner state

The model uses patterns learned across many examples.

Practice across images, noise levels, and timesteps

Repeated comparison adjusts the model so future denoising predictions improve.

01

Choose

Select a training image and a timestep.

02

Corrupt

Add the matching amount of noise.

03

Predict

Estimate the noise or cleaner direction.

04

Learn

Adjust internal parameters using the error.

Patterns are reused; pixels are newly formed

Denoising creates a new trajectory.

The system starts from a random state and follows learned visual tendencies. It is not simply pulling one complete picture from a hidden album.

Learned patterns $\neq$ stored gallery

Constructing the training challenge

The forward process: adding noise

A controlled schedule gradually replaces image signal with randomness, producing examples at many difficulty levels.



Noise is random variation added to visual values

In a simplified picture, noise disturbs local brightness and color so the original structure becomes harder to detect.

Signal

Organized visual pattern

Edges, colors, and relationships support a recognizable scene.

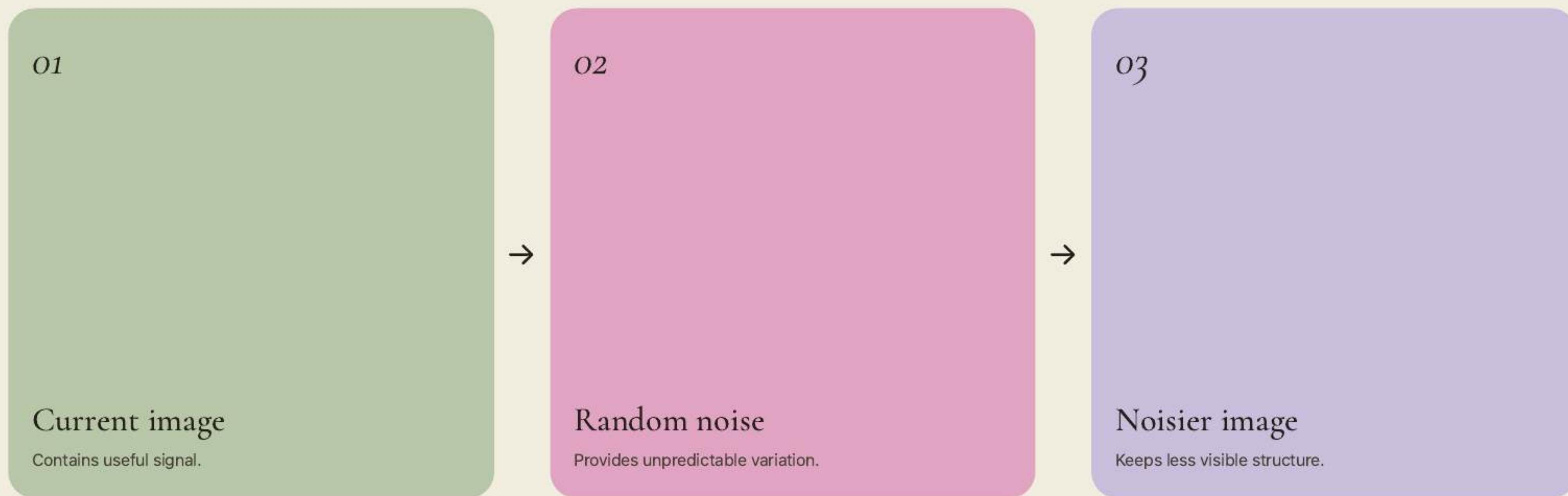
Noise

Random visual variation

Unpredictable changes interfere with those relationships.

Mix a little randomness into the current image

The amount of added noise depends on the chosen schedule and timestep.



Early, middle, or late in the process

Different timesteps contain different amounts of signal.

An early noisy image may preserve edges. A late one may look almost random. The model must learn across the entire range.

timestep $\approx$ how noisy is this example?

Forward diffusion

Recognizable structure becomes weaker

The transition is designed so the model encounters many intermediate states—not only clean and fully noisy endpoints.

mostly signal



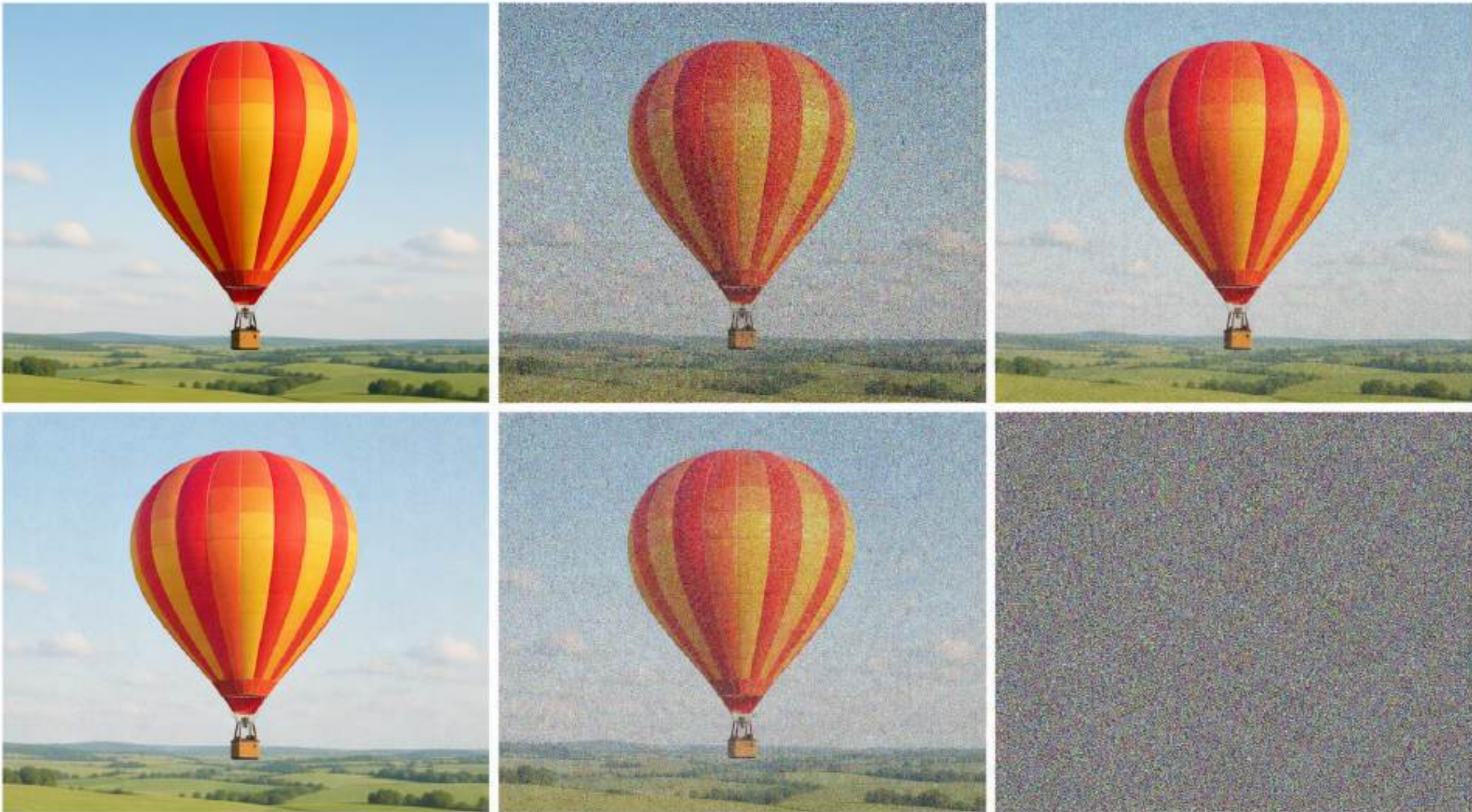
mostly noise

The exact schedule is a technical design choice; the high-level idea is gradual corruption.

Six shuffled images · one process

Put the noisy frames in order

Arrange the balloon frames from clearest to most random. Then explain which visual clues guided your decision.



Build the sequence

- 01 When does the silhouette disappear?
.....
- 02 When does color stop helping?
.....
- 03 Which two stages are hardest to place?
.....

Look for signal strength—not just how attractive the image looks.

Create questions with known answers

Adding noise gives the model something measurable to learn.

Because training began with a known image, the system can evaluate how well its prediction points toward a cleaner state.

Known clean image + known corruption = supervised practice signal

Generation runs the story in the useful direction

Iterative denoising

Start from random noise, predict a small correction, update the image, and repeat until a coherent result appears.



A fresh random seed

Generation begins with noise.

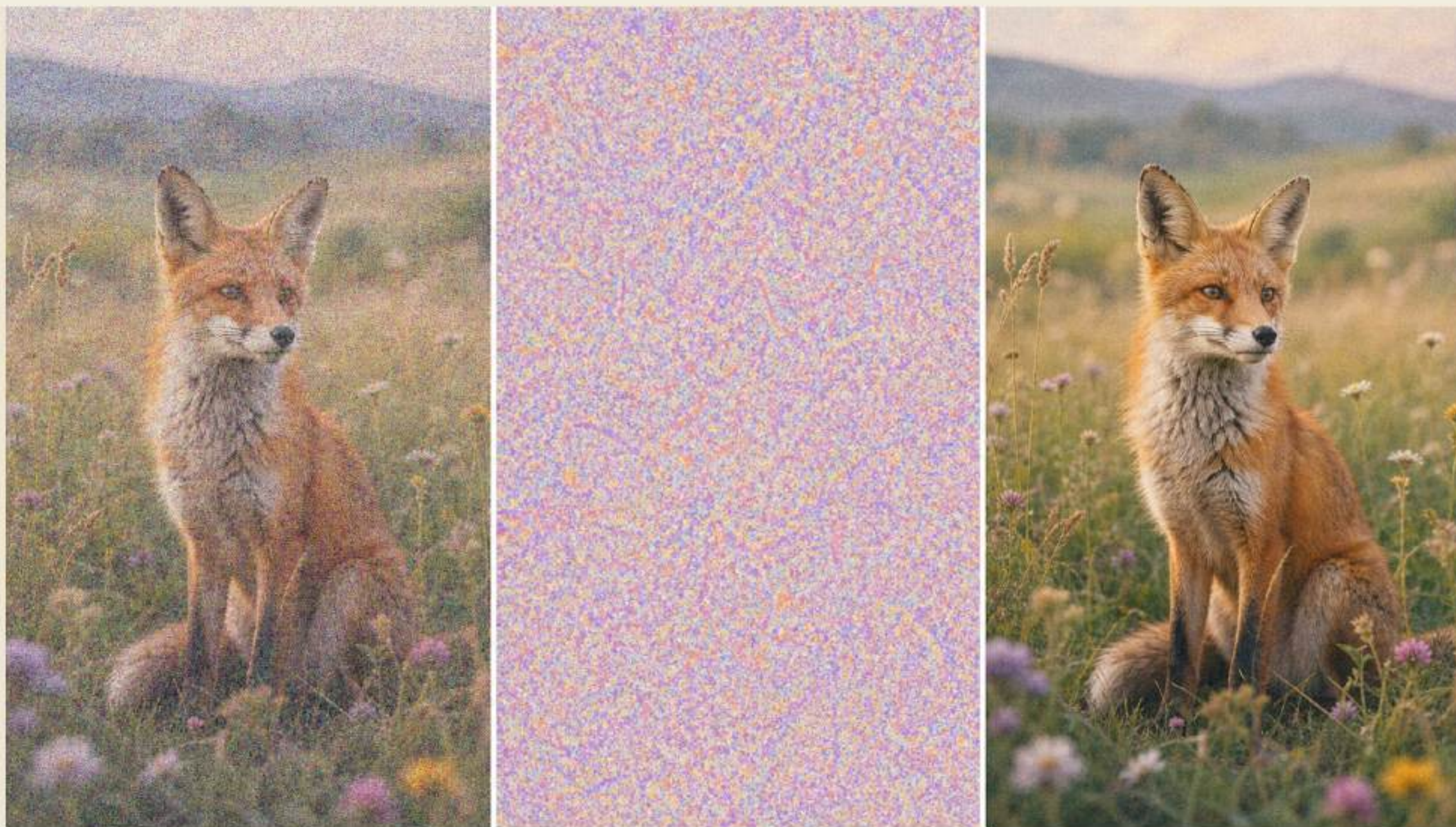
At the start, there is no visible fox, robot, or landscape. The initial random values provide a starting point for the denoising trajectory.

Different starting noise can lead to different outputs.

Current state → *predicted noise* → *cleaner direction*

One prediction points toward a cleaner state

The middle panel visualizes estimated noise. Real systems apply many updates; this triptych enlarges the idea of one direction, not a literal one-step jump to the final image.



Predict, update, and repeat

The same broad cycle runs across many timesteps.



Six moments in one denoising path

Coarse structure arrives before fine detail

The model first establishes broad composition and subject shape. Texture and small features become clearer later.



Large-scale structure must settle first

Early updates strongly influence the overall arrangement.

01

Composition

Where major regions and subjects appear.

02

Silhouette

The broad shape of the main subject.

03

Color blocks

Large areas of light, dark, warm, and cool.

Fine details refine the established scene

Later steps usually have less freedom to reorganize the entire composition.

01

Texture

Fur, fabric, grass, paper, or metal.

02

Edges

Sharper boundaries and smaller shapes.

03

Lighting

Local highlights and shadows.

04

Details

Small marks that support realism.

Same prompt · four starting seeds

One instruction can lead to several valid images

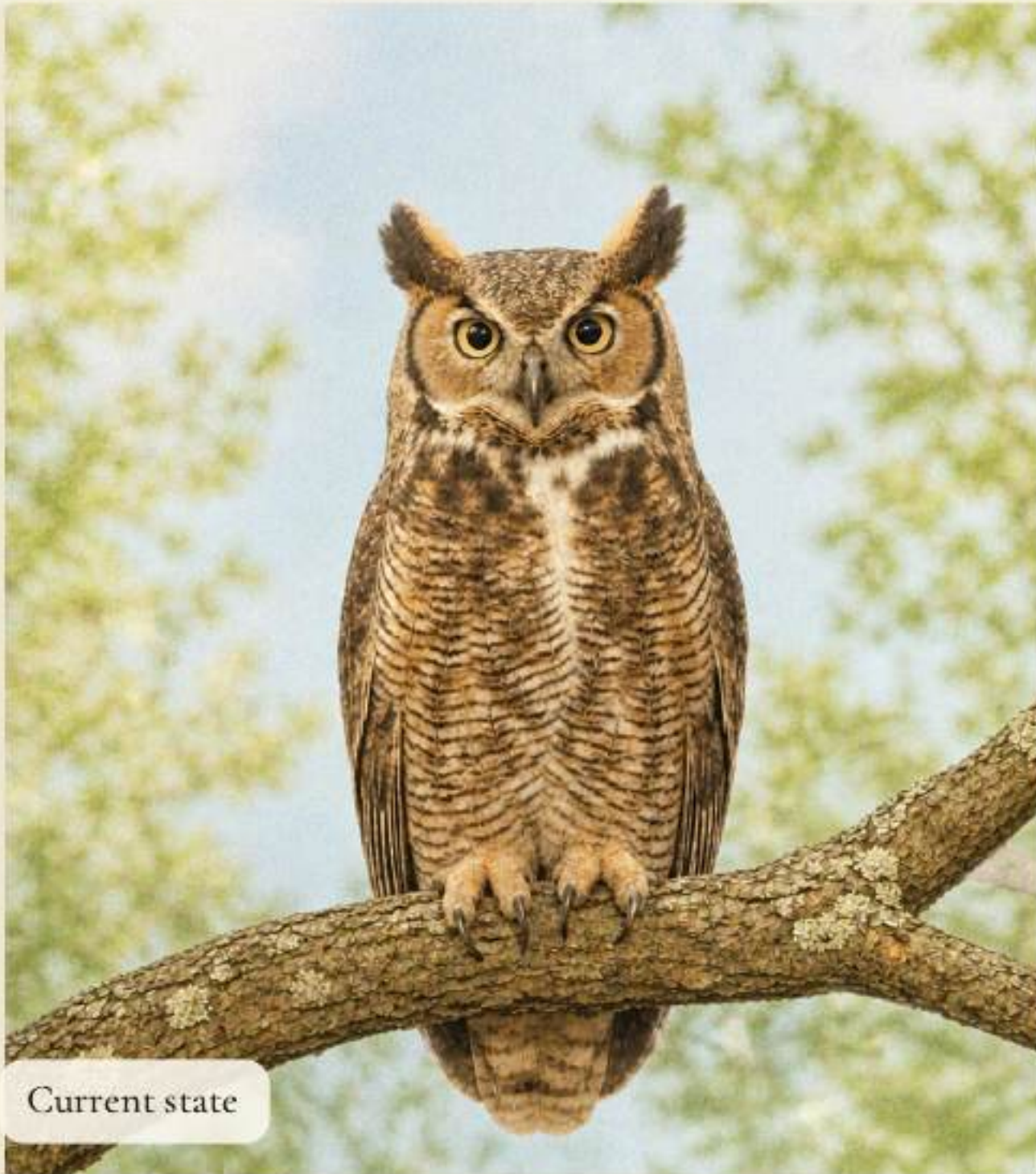
Every result follows the greenhouse-robot prompt, yet the camera, robot design, plants, and composition differ because generation began from different noise.



Observe → predict → reveal

What might become clearer next?

Study the current owl image. Decide what is already stable and what may sharpen during the next small denoising update.



Predict first

Click to reveal the next stage

Already visible

- Owl and branch positions
- Broad silhouette
- Main color regions

Still uncertain

- Fine feather groups
- Bark texture
- Small highlights

Reveal next step

Composition → structure → texture → detail

Denoising in a more efficient representation

Latent diffusion

Instead of repeating every operation across full-resolution pixels, many systems compress images, denoise in latent space, and decode at the end.



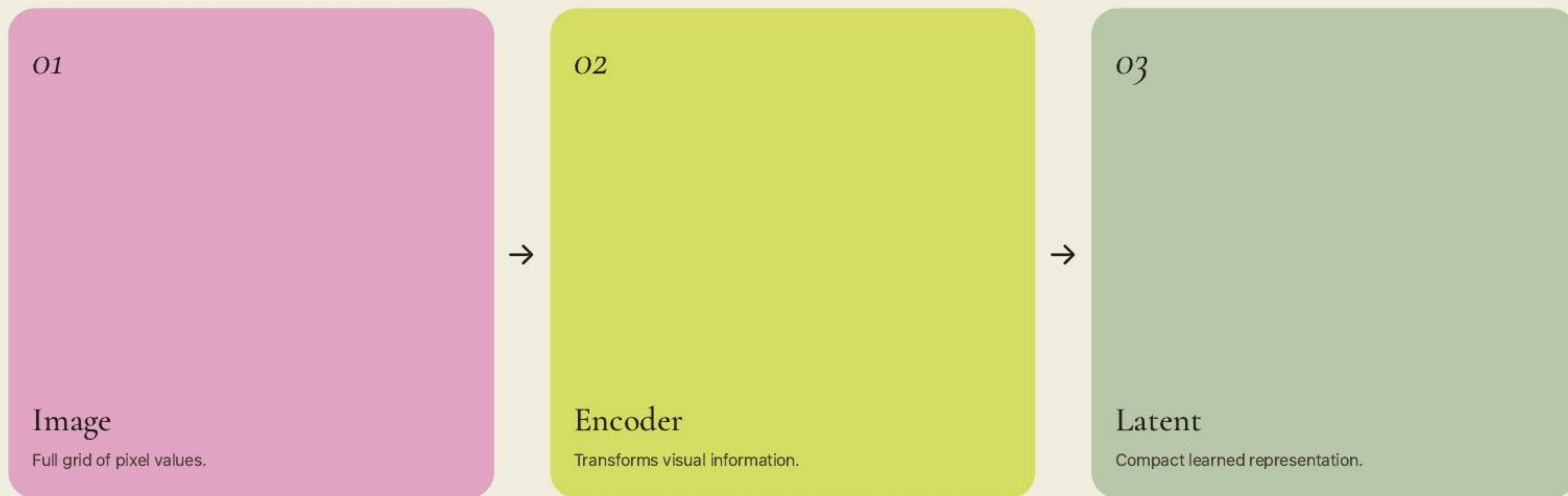
Millions of values · many repeated steps

High-resolution denoising requires substantial computation.

Every extra pixel adds values that must be processed across multiple timesteps. A compact representation can reduce this burden.

Compress the image before diffusion

The representation aims to preserve important visual structure while using a smaller working space.



Denoising can happen without touching every final pixel

The model modifies a compressed representation that still carries useful visual patterns.

Pixel space

Detailed but large

Directly represents final image locations and channels.

Latent space

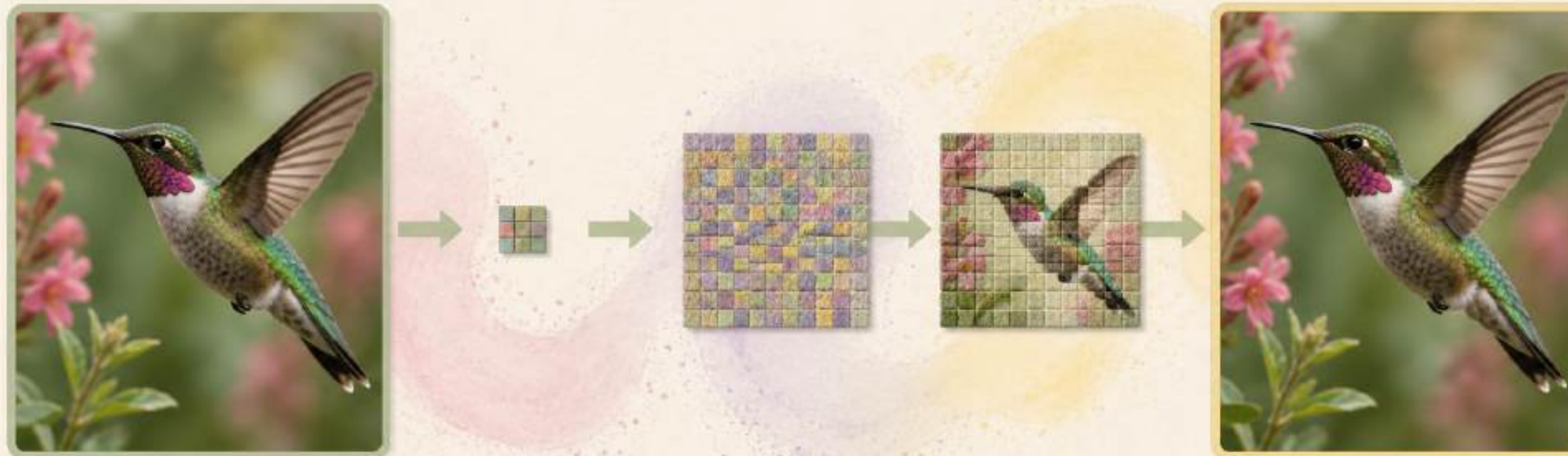
Compact but meaningful

Represents learned visual information in a reduced form.

Image → *compact latent* → *refined latent* → *pixels*

The model works on a smaller visual canvas

A compact representation is easier to process repeatedly. After denoising, the decoder reconstructs a visible image with color and detail.



From compact representation to image

A final latent must be decoded before we can view it.

The decoder reconstructs pixel-level color and detail from the generated representation.

Latent representation → decoder → image

Efficiency can come with information loss

A useful latent keeps important structure, but no compression is a perfect copy of every pixel.

Benefit

Faster, smaller working space

The diffusion model can spend computation on a compact representation.

Trade-off

Some exact detail may be harder to preserve

The decoder must reconstruct visible pixels from compressed information.

The generation engine now has four stages

Text guidance will join this process in the next section.

01

Noise

Begin with uncertainty.

02

Predict

Estimate a cleaner direction.

03

Repeat

Build structure across timesteps.

04

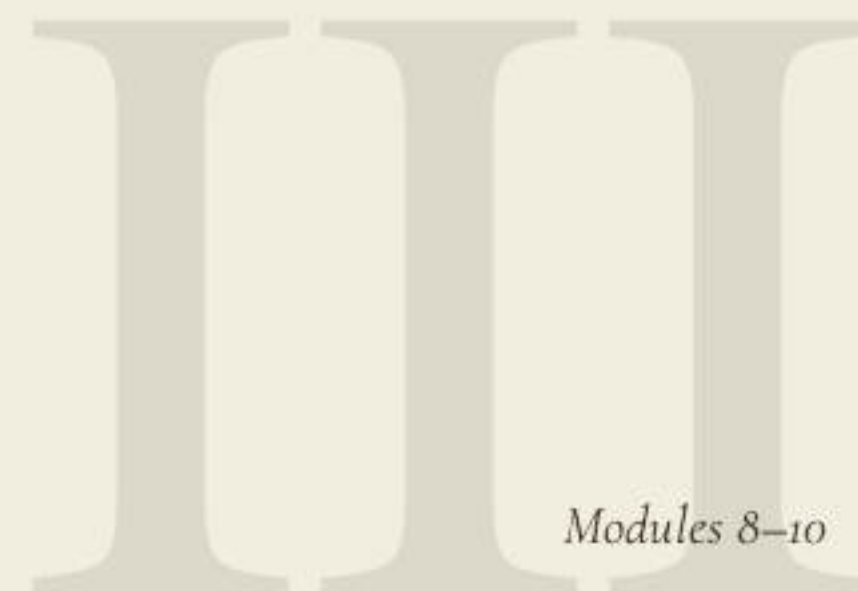
Decode

Return to visible pixels.

Words influence the visual path

How text guides an image

A prompt becomes a numerical representation that conditions denoising.
Specificity changes which visual possibilities receive stronger support.



Connecting language and vision

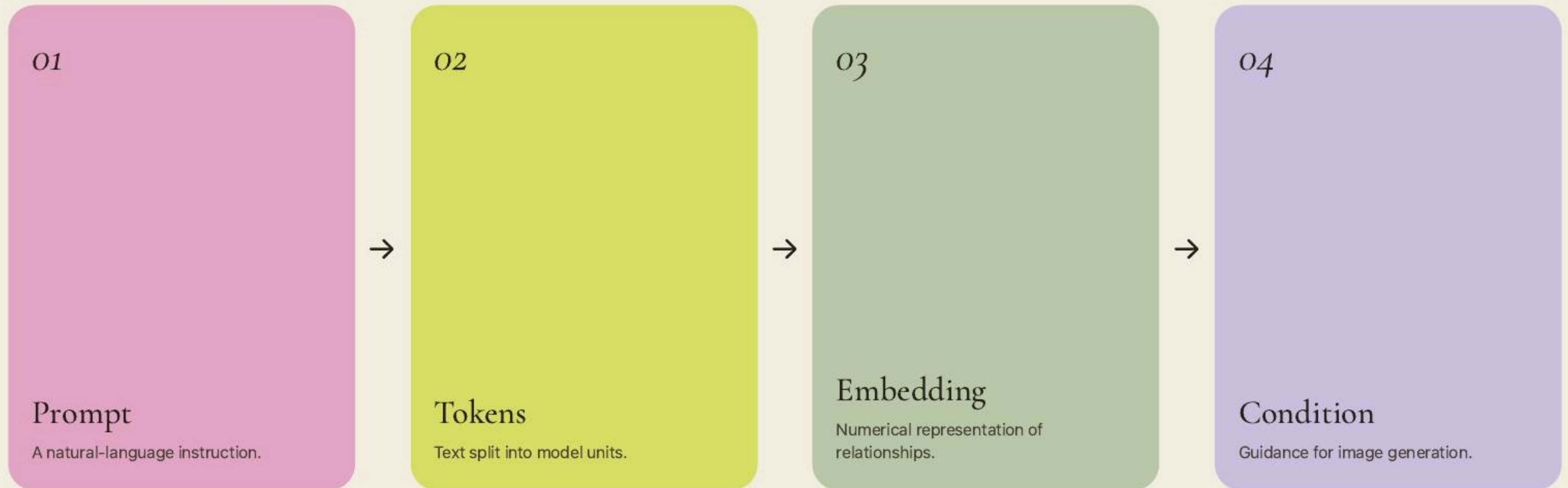
Text embeddings & conditioning

The system represents the prompt numerically, then uses that representation to guide many denoising predictions.



Text must become a form the image model can use

The early steps connect directly to ideas from Day 1.



Meaning becomes numerical pattern

Related descriptions can produce related representations.

Words such as "puppy" and "dog" may share useful relationships, while modifiers such as "watercolor" can shift the desired visual style.

Embedding = learned numerical representation

The model uses relationships— not a dictionary lookup

A prompt representation combines words with their surrounding context.

Individual word

“bank” is ambiguous

It could describe money, a building, or the side of a river.

Contextual phrase

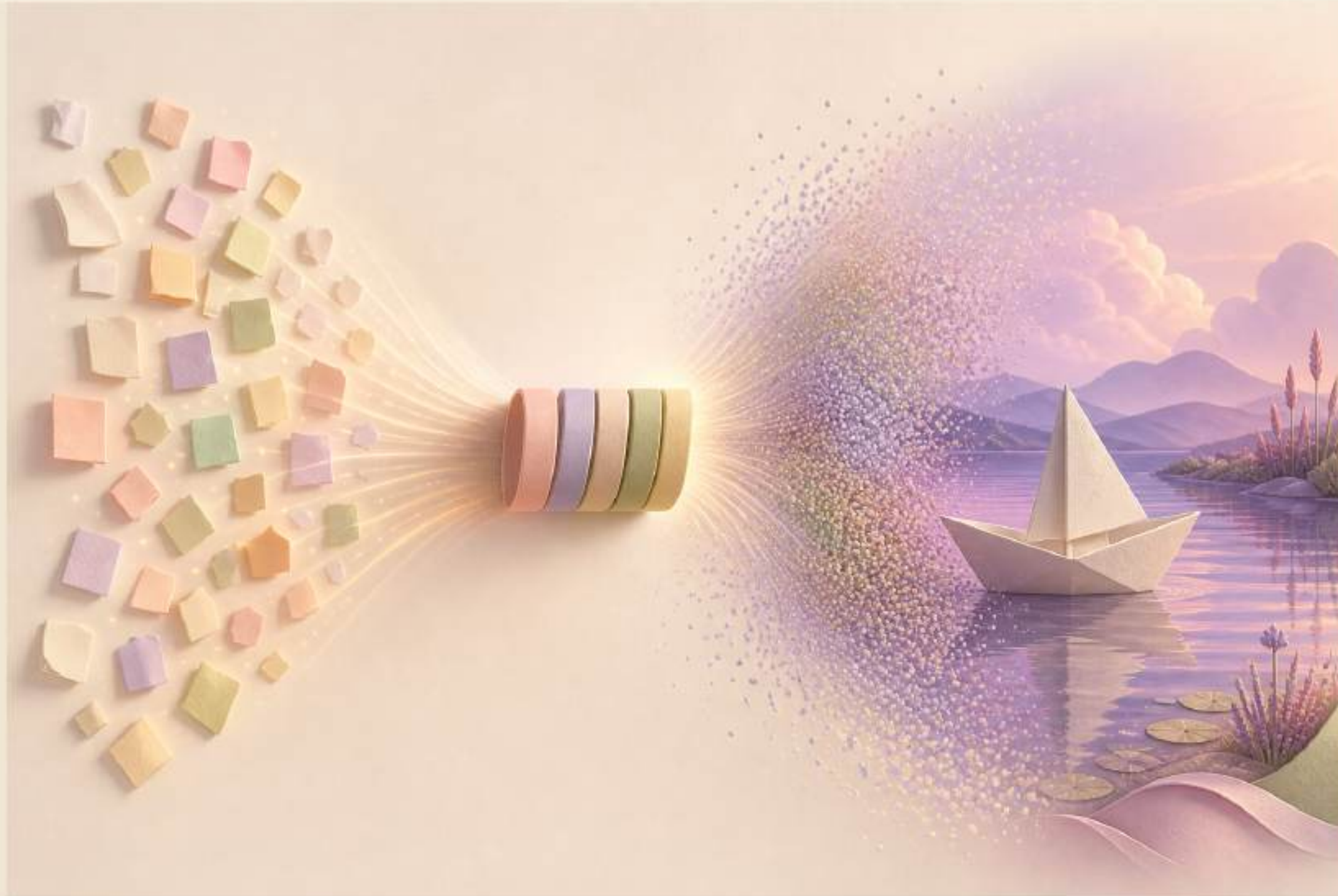
“A boat near the river
bank”

Nearby words support the landscape meaning.

Words guide the path without becoming pixels directly

The prompt shapes which image becomes more likely

The text representation interacts with the generation process across many steps, encouraging visual features that match the instruction.



Not one instruction at the beginning

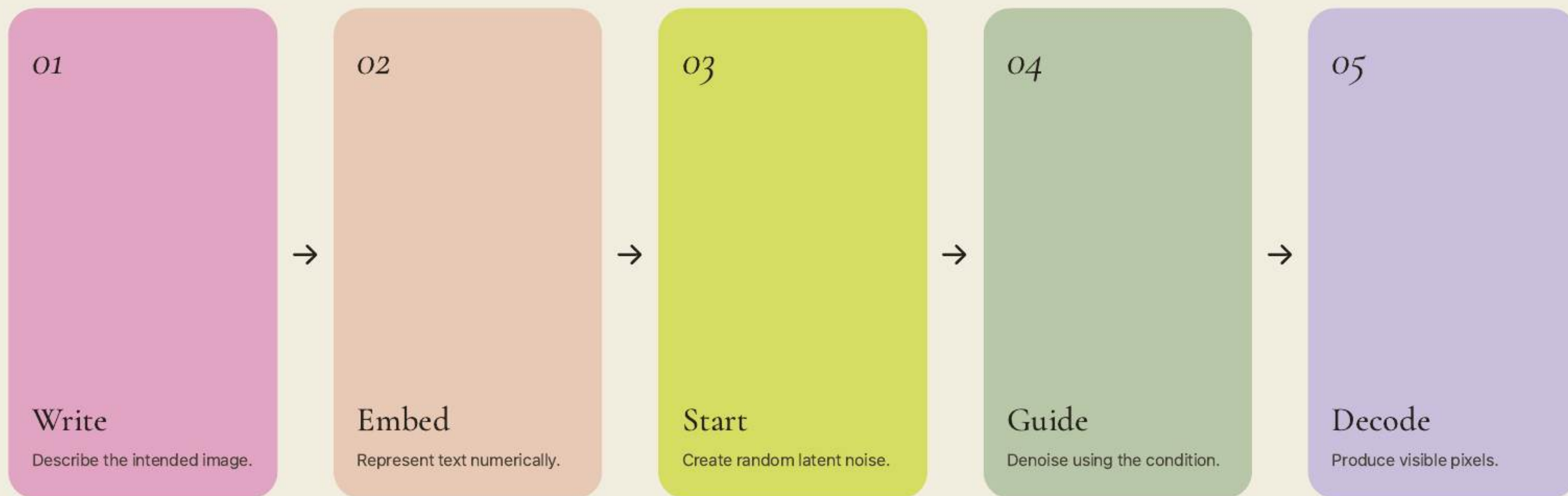
Text can influence many denoising steps.

At each stage, the model compares the current representation with the conditioning information and predicts an update that better fits both.

Current latent + timestep + text condition → update

Prompt to pixels in six high-level stages

Different systems vary, but this mental model captures a common latent-diffusion workflow.



Weak → balanced → excessive

More prompt pressure is not always better

Weak guidance may lose the subject; balanced guidance can be coherent; excessive pressure may create repetition, harsh color, or unnatural artifacts.



Turning intention into useful language

Prompt specificity & style

A good prompt communicates the decisions that matter most: subject, setting, composition, lighting, mood, and visual treatment.

Add decisions that matter

Useful detail narrows the space of plausible images.

A broad prompt leaves many choices to the model. A specific prompt communicates which choices should remain flexible and which should be controlled.

Specificity reduces ambiguity; it does not guarantee perfection.

Compare four prompt levels

Specific instructions shape the visual result

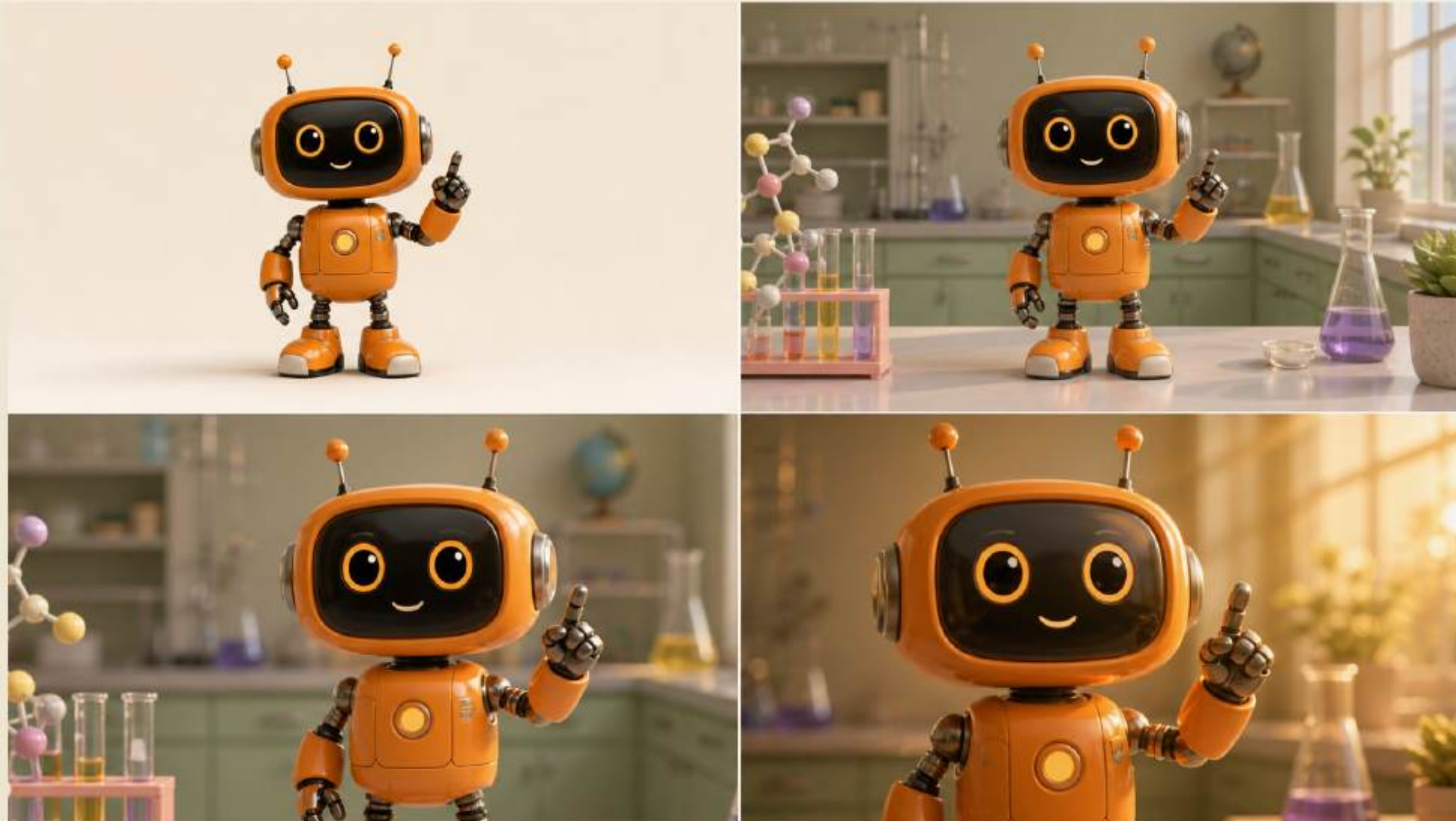
The robot stays recognizable while setting, activity, composition, and lighting become more intentional.



Subject → setting → framing → lighting

Each useful phrase should create a visible change

The robot stays recognizable while the prompt adds a science lab, a close-up composition, and warm golden lighting.



Describe where the scene happens

Setting influences background objects, scale, lighting, and the story implied by the image.

No setting

The model chooses the world

The subject might appear in a studio, street, forest, or empty space.

Clear setting

“In a bright school science lab”

The scene gains useful environmental constraints.

Tell the model how the scene should be framed

Composition describes placement and viewpoint, not just the objects present.

Framing

Wide, close-up, overhead,
low angle

Changes how much context and detail the viewer sees.

Placement

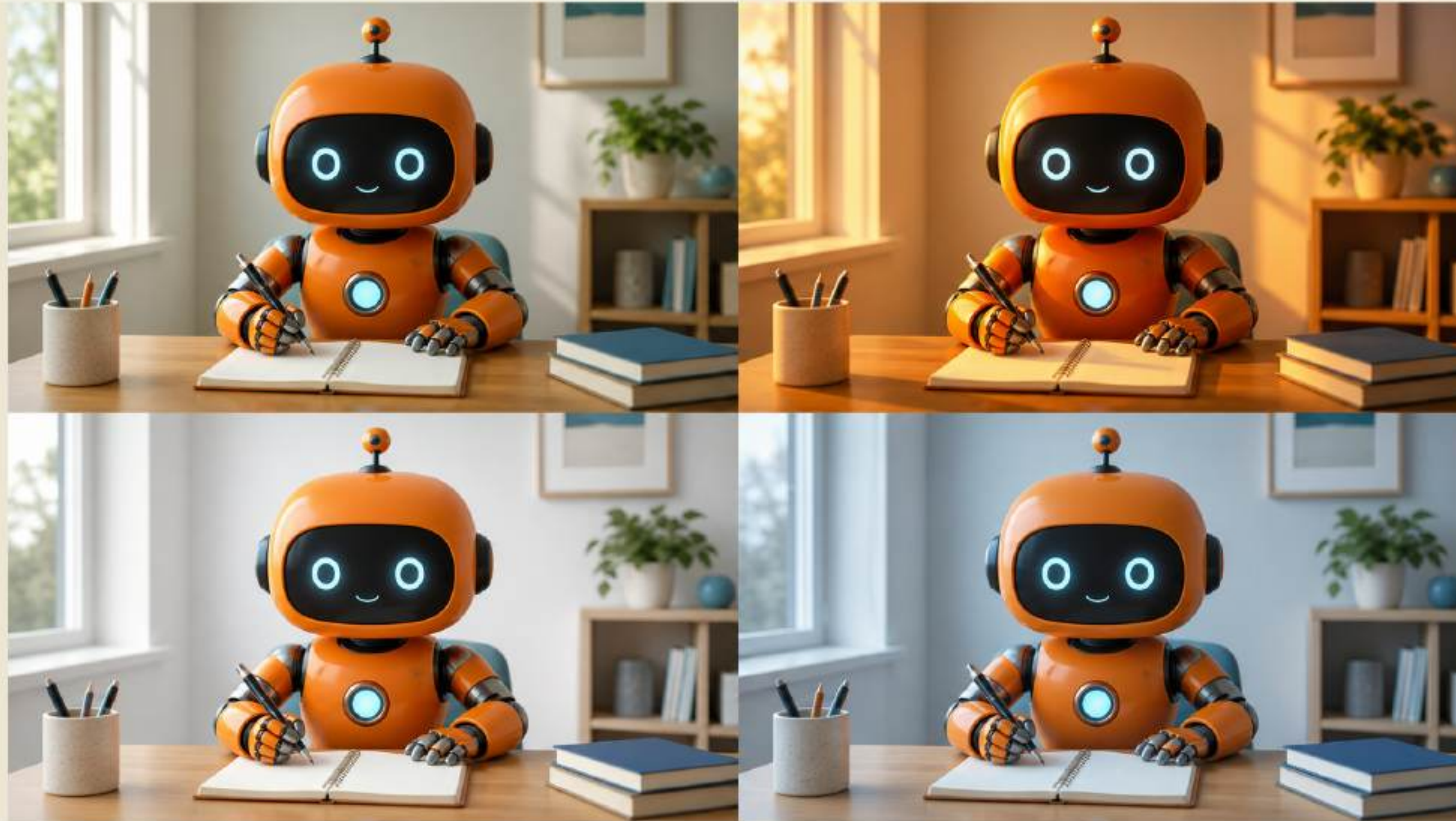
Centered, right third,
negative space

Helps reserve room and organize visual attention.

Same subject · same camera · different light

Lighting changes how a scene feels

Soft daylight, golden hour, studio light, and overcast light alter contrast, color temperature, shadows, and emotional tone without changing the scene.



Same subject · different visual language

Medium and style influence texture, edges, and color.

Words such as “watercolor,” “clay animation,” or “editorial photograph” can guide how the subject is rendered.

Style describes appearance—not the factual truth of the scene.

One fox · four visual treatments

A modifier can reshape the entire image

The subject and composition remain similar, while texture, line quality, depth, and realism change dramatically.



Specificity should serve the goal

Extra words can conflict, distract, or add noise.

A strong prompt prioritizes the few decisions that matter most. Too many competing instructions can make the target less clear.

Relevant detail > maximum detail



Prompting works best as an observation-and-revision process.

Generate → inspect → revise

Improve one prompt as a team

Begin with a broad idea. Add subject, setting, composition, lighting, and style only when each decision supports the goal.

Change one major variable at a time so you can learn what caused the difference.

Polished images still need human judgment

Coherence & responsible use

A good-looking image should also make sense: objects, lighting, scale, perspective, purpose, and provenance all deserve inspection.

The parts of the scene should agree

Coherence is the feeling that the image follows one consistent visual world.

01

Geometry

Shapes and perspective fit together.

02

Lighting

Highlights and shadows share a plausible source.

03

Identity

Objects and people remain consistent.

04

Relationships

Positions, scale, and contact make sense.

Look twice

Can you find five inconsistencies?

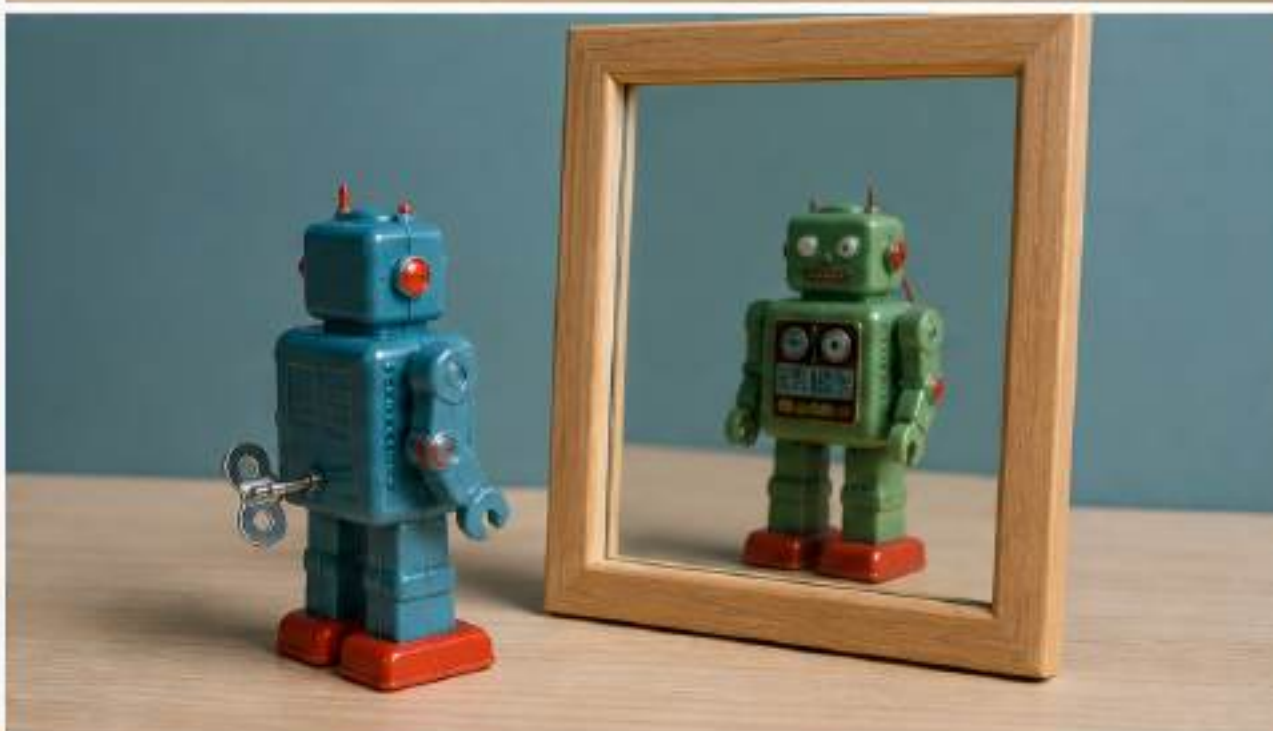
Inspect geometry, shadows, reflections, object structure, and impossible overlaps. A polished first impression is not enough.



Inspect hands · counts · light · reflections

Polished details can still disagree

Look for an extra finger, a quantity mismatch, a shadow pointing the wrong way, and a reflection showing a different robot.



Creative freedom and factual evidence are different goals

The same visual quality can be acceptable for one task and misleading for another.

Creative concept

Exploration can welcome invention

A fictional creature or mood board may benefit from surprise.

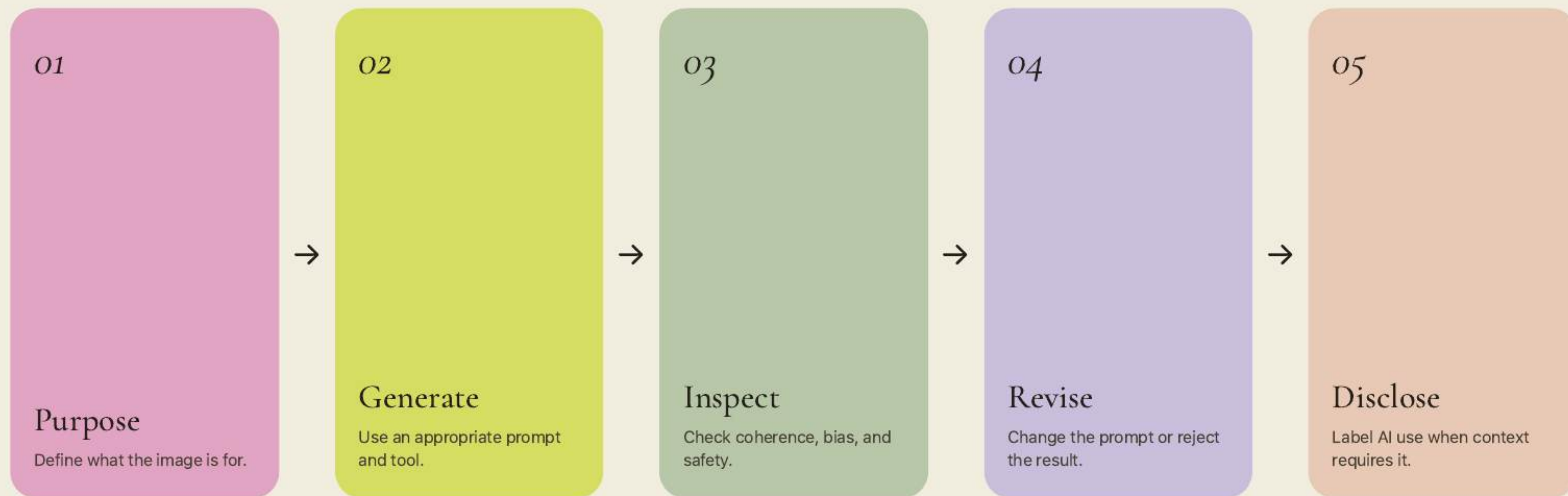
Evidence or news

Real-world claims require verification

A generated image should not be presented as proof that an event occurred.

Create, inspect, revise, and label

Human judgment remains part of the process from prompt to publication.



Keep the whole mental model

Words guide. Noise changes. Images emerge.

A prompt becomes a representation. That condition guides repeated denoising in a visual space.

A decoder produces pixels—and a thoughtful human decides whether the result is useful, coherent, and responsible.

Prompt → embedding → latent noise → guided denoising → image → human review